

Mass resolution, observed upper limits, and a one-sided width nuisance

Standalone continuation of the v5.3 resolution study

Version 5.3.1 — 10 September 2026

1 Study and main results

The unscaled Monte Carlo (MC) resolution is used as the starting point for each dataset. Four width scenarios follow the documented correction toward the fully scaled resolution. The full 2015 and 2016 samples and the native 2021 10% sample are scanned separately and combined with one shared coupling. This continuation extends 2015 to 100 MeV and adds an actual continuous, one-sided resolution nuisance for the individual datasets on a fixed observed experiment.

The two comparisons answer different questions. Repeating the moving-window GP analysis measures the response of the whole fitting procedure to resolution. Profiling the width on fixed bins and a fixed background constraint measures its effect within one likelihood. The latter uses an explicitly assumed calibration constraint; it is not a measured resolution-uncertainty determination.

Dataset	Mass [MeV]	Nodes	Scenario scaled/MC	Profiled/fixed MC
			median [min, max]	median
2015	19–100	82	1.3 [0.771, 2.77]	1.15
2016	39–180	142	1.8 [0.42, 7.76]	1.56
2021	50–250	201	1.27 [1.1, 1.95]	1.08
Combination	19–250	232	1.31 [0.733, 2.77]	–

The scenario column compares the moving-window analyses. The nuisance column compares the assumed Gaussian width profile with the MC-width control on identical fixed bins and the same GP constraint. These are medians of pointwise observed 90% CL_s limit ratios, rather than expected-limit ensembles. Local excesses and deficits can produce nonmonotonic changes and move the minimum p-value in mass. For 2021, the 25% width increase gives a median 27.1% increase in the observed limit and 26.0% increase in the fitted full-template yield error. Narrower resolution therefore reduces the fitted error in this comparison; that reduction does not specify how a calibration nuisance propagates into an interval.

With the one-sided Gaussian constraint, the 2021 profiled limit is instead 7.5% higher than the comparable fixed-experiment MC limit in median, or 9.2% above the moving-window MC result. The difference between these comparisons comes from fixing the GP exclusion and fit bins before profiling the width.

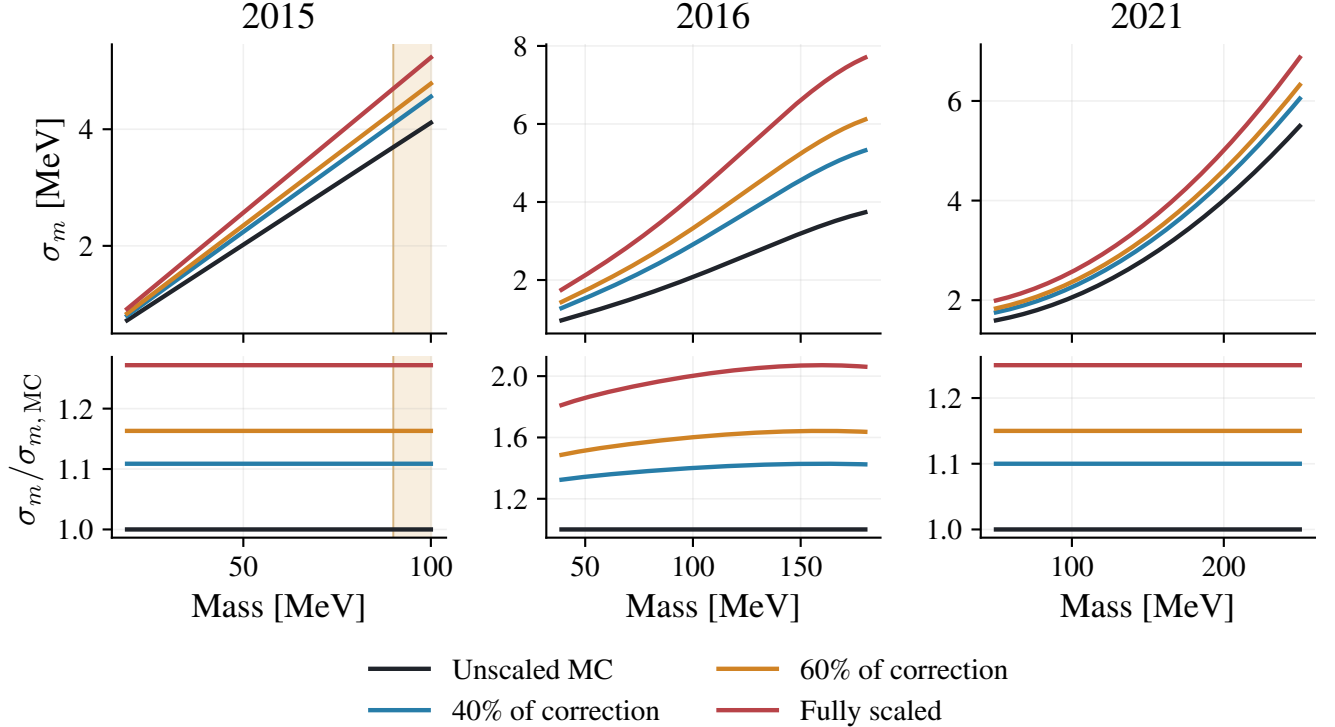
No expected-limit bands, new pseudoexperiments or global p-values are introduced. All results remain conditional on the stated background and resolution models. The 2015 extension has not acquired new support or coverage qualification.

2 Mass-resolution inputs and scenario construction

For dataset d , the scenarios interpolate between the source MC and fully scaled curves:

$$\sigma_{d,t}(m) = \sigma_{d,\text{MC}}(m) + t[\sigma_{d,\text{scaled}}(m) - \sigma_{d,\text{MC}}(m)], \quad t \in \{0, 0.4, 0.6, 1\}.$$

The 2021 multipliers are 1.00, 1.10, 1.15 and 1.25. The 2015 multipliers are approximately 1.000, 1.109, 1.163 and 1.272. The 2016 correction is mass dependent, with its fully scaled/MC ratio ranging from about 1.811 to 2.071 on this scan.

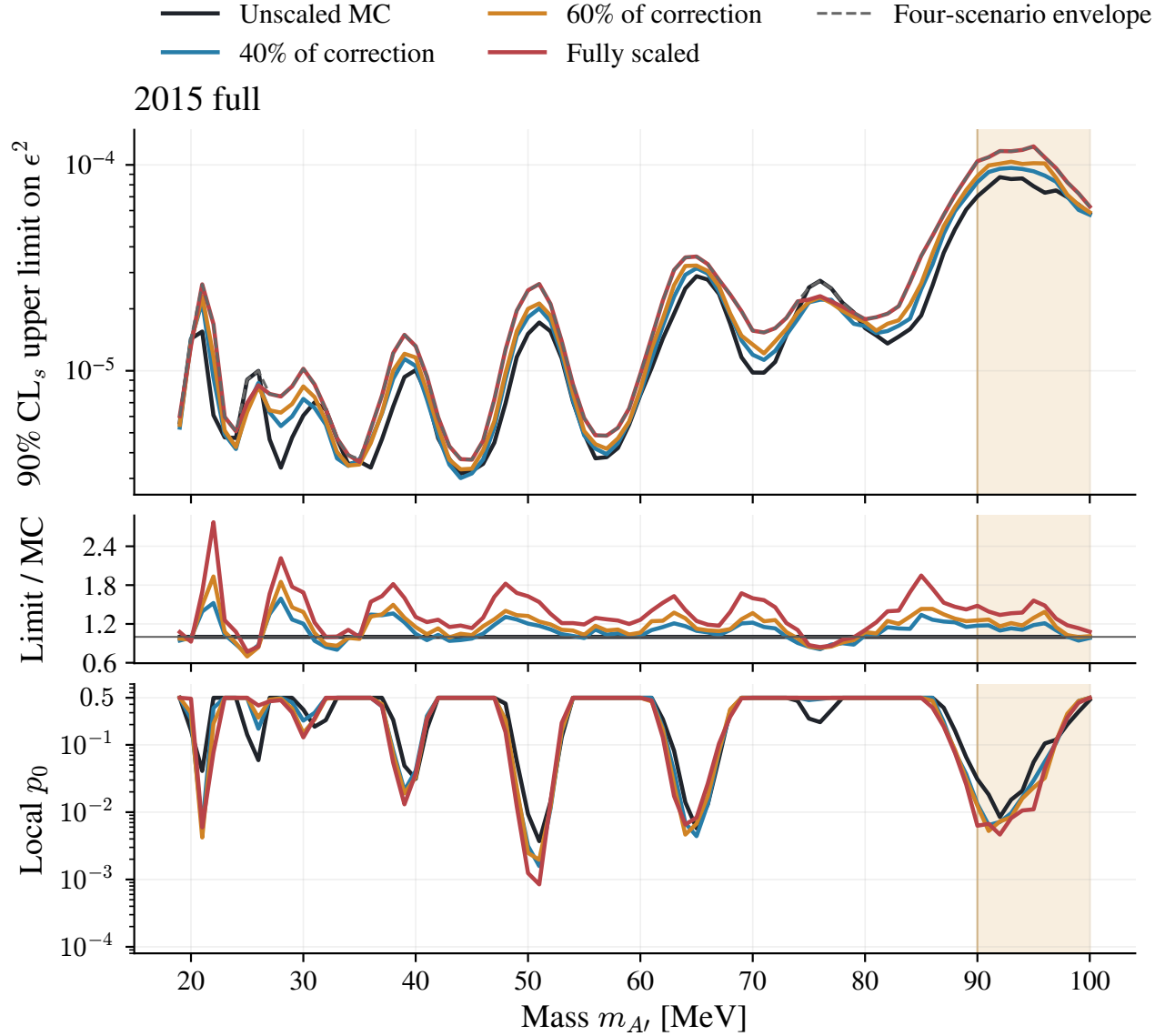


Mass-resolution curves and their ratios to MC. Orange shading marks the exploratory 2015 extension from 90 to 100 MeV. The 2016 curves are interpolated pointwise; a single constant scale would not reproduce their source polynomials.

Each scenario rebuilds the moving $\pm 2.25\sigma$ extraction and GP training exclusion. GP hyperparameters are selected using sideband marginal likelihood from 12 seeded restarts, an admissible reviewed-kernel seed and its local optimization. This uniform sideband-only rule does not guarantee the global optimum. Signal templates, GP covariance and the native observed prompt density within $\pm 1.64\sigma$ are recomputed.

The upper-limit fit profiles the Poisson likelihood and Gaussian GP background constraint. Observed and background-Asimov likelihood ratios are calculated separately before applying the bounded asymptotic CL_s tails. The limit solves $\text{CL}_s = 0.10$. Displayed ϵ^2 includes the inherited dimuon correction above threshold, applied identically to every scenario. It cancels in matched ratios and does not affect p_0 .

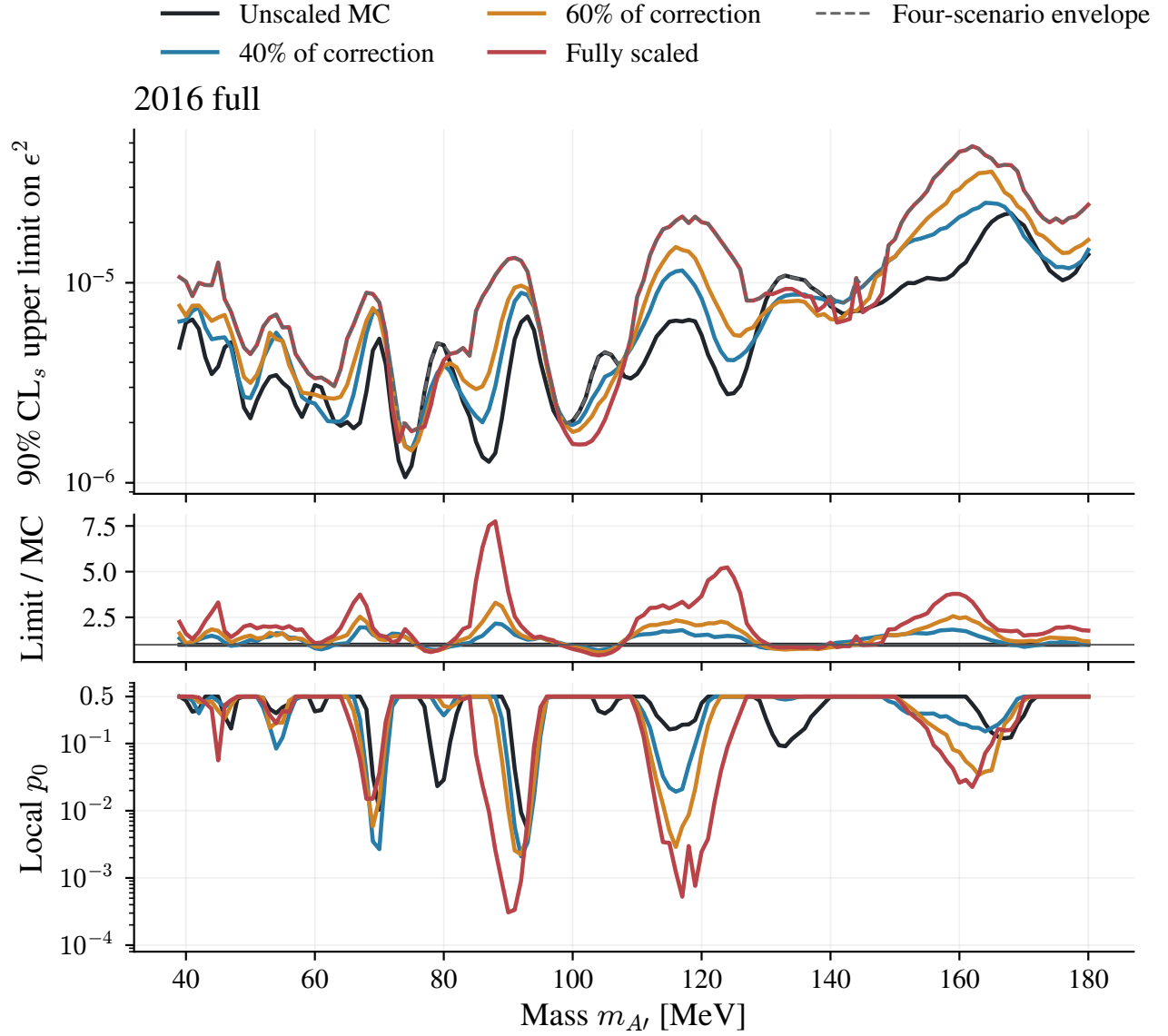
3 2015: observed limits and local p-values



Observed limits (top), ratios to the unscaled MC scenario (middle), and fixed-mass asymptotic p_0 on a logarithmic probability axis (bottom). The dashed envelope is the maximum over the four tested widths; it has no one-standard-deviation interpretation. Values at $p_0 = 0.5$ correspond to no positive fitted excess.

The fully scaled/MC limit ratio has median 1.3 and ranges from 0.771 to 2.77. The minimum MC-scenario p_0 is 3.7×10^{-3} at 51 MeV; the fully scaled minimum is 8.5×10^{-4} at 51 MeV. These minima are local references, not probabilities for the full mass-and-width search. Orange shading identifies 90–100 MeV, where the 2015 scan is exploratory.

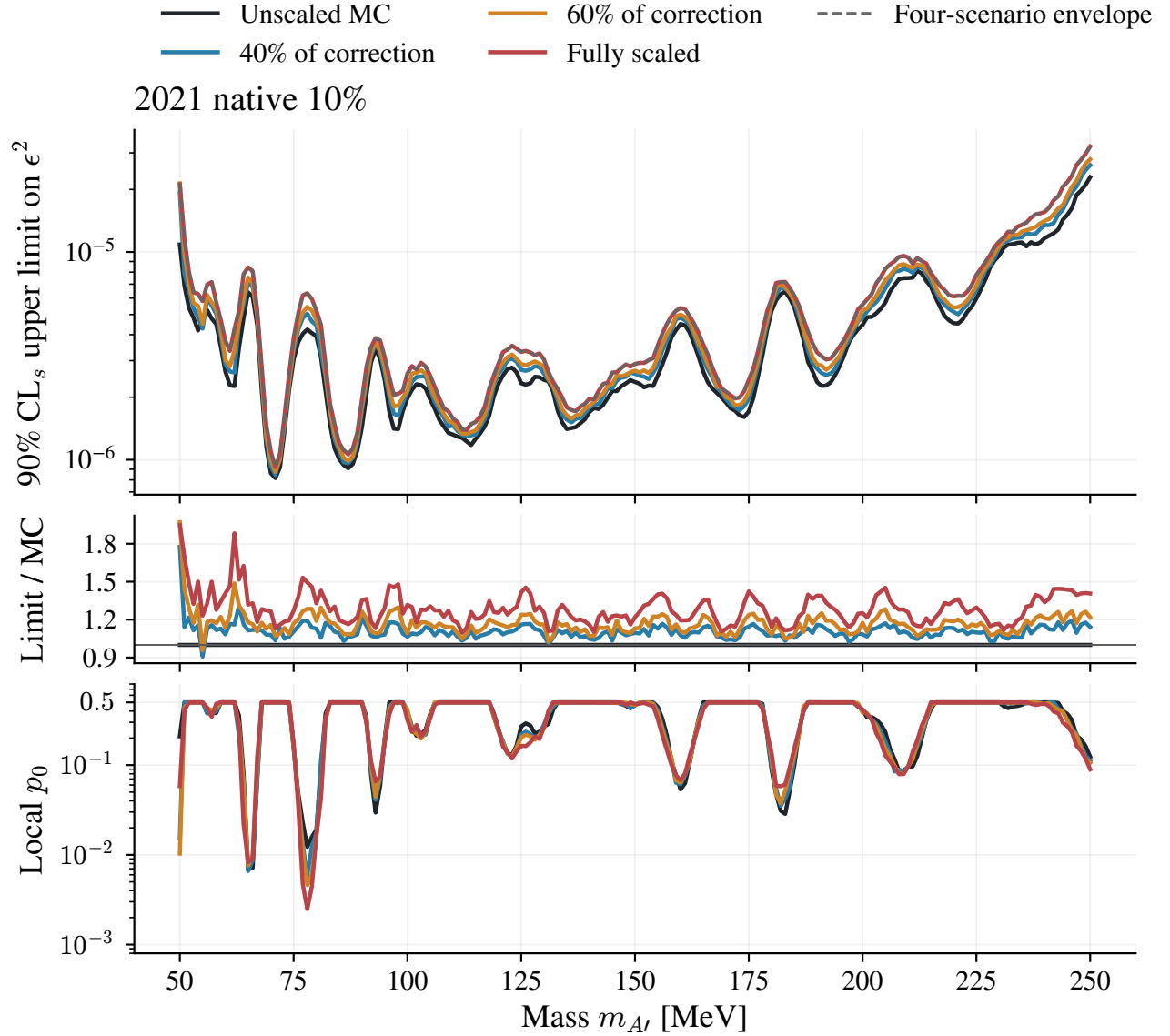
4 2016: observed limits and local p-values



Observed limits (top), ratios to the unscaled MC scenario (middle), and fixed-mass asymptotic p_0 on a logarithmic probability axis (bottom). The dashed envelope is the maximum over the four tested widths; it has no one-standard-deviation interpretation. Values at $p_0 = 0.5$ correspond to no positive fitted excess.

The fully scaled/MC limit ratio has median 1.8 and ranges from 0.42 to 7.76. The minimum MC-scenario p_0 is 5.3×10^{-3} at 93 MeV; the fully scaled minimum is 3.1×10^{-4} at 90 MeV. These minima are local references, not probabilities for the full mass-and-width search.

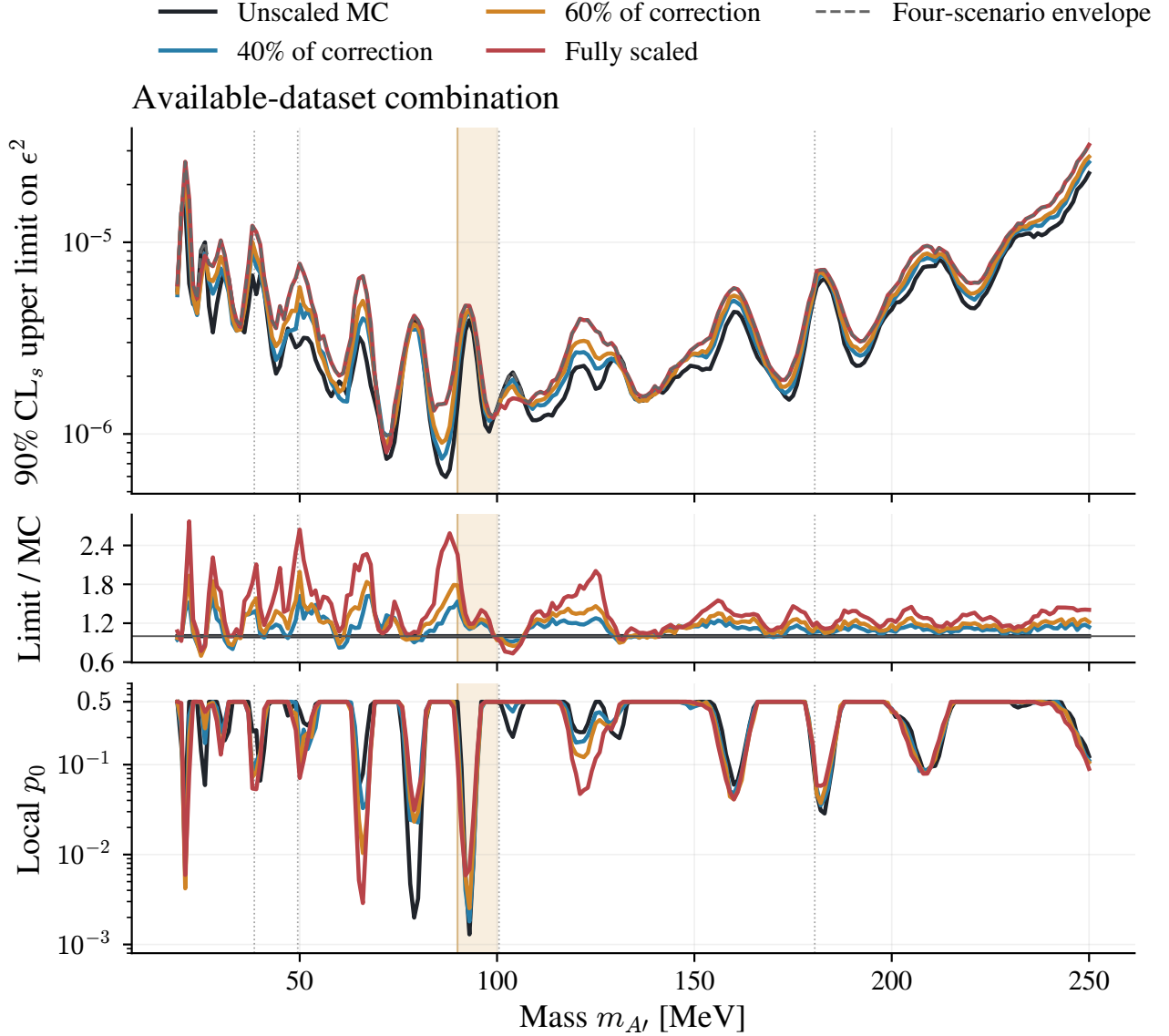
5 2021: observed limits and local p-values



Observed limits (top), ratios to the unscaled MC scenario (middle), and fixed-mass asymptotic p_0 on a logarithmic probability axis (bottom). The dashed envelope is the maximum over the four tested widths; it has no one-standard-deviation interpretation. Values at $p_0 = 0.5$ correspond to no positive fitted excess.

The fully scaled/MC limit ratio has median 1.27 and ranges from 1.1 to 1.95. The minimum MC-scenario p_0 is 6.9×10^{-3} at 65 MeV; the fully scaled minimum is 2.5×10^{-3} at 78 MeV. These minima are local references, not probabilities for the full mass-and-width search.

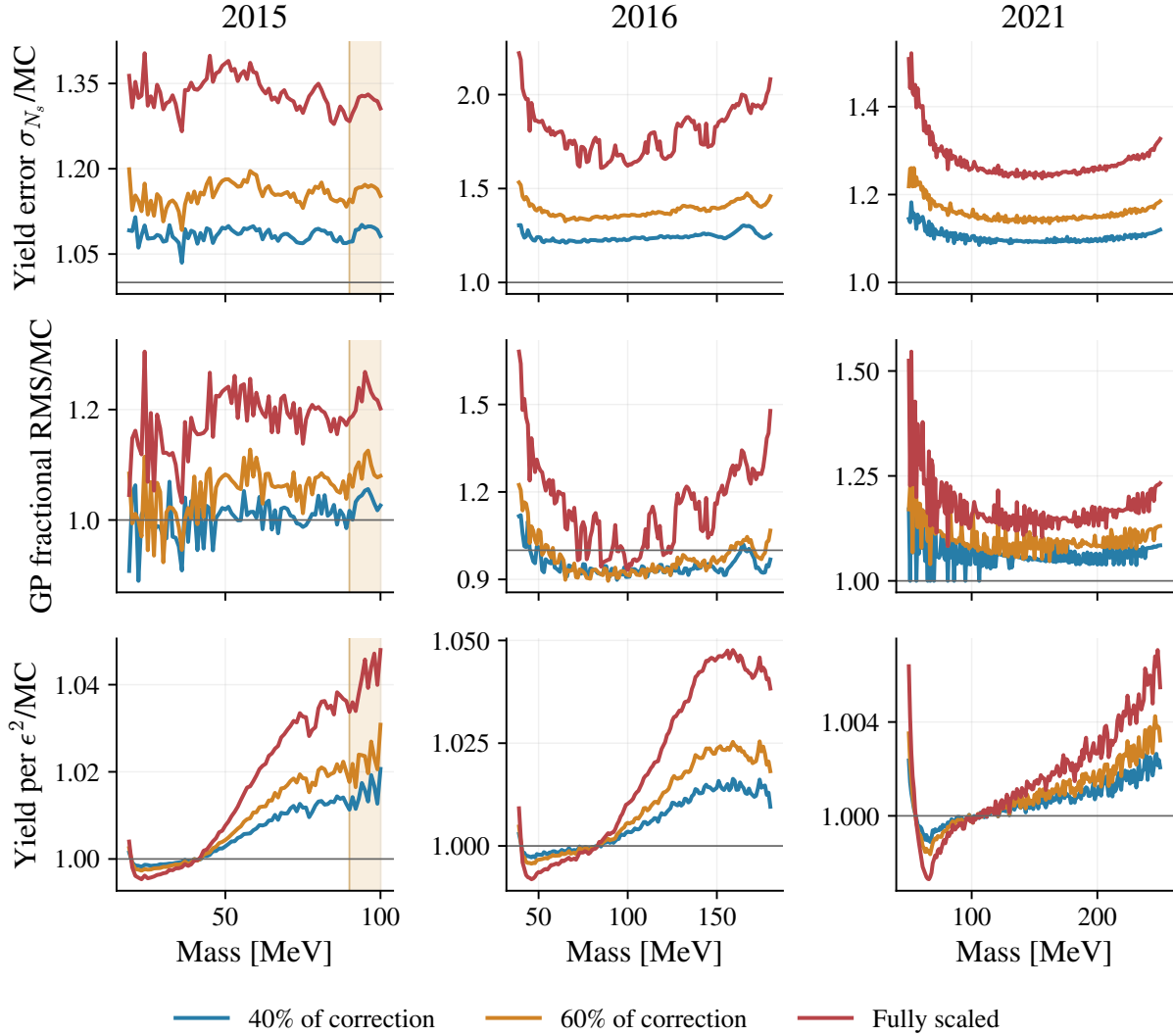
6 Combination: observed limits and local p-values



Observed limits (top), ratios to the unscaled MC scenario (middle), and fixed-mass asymptotic p_0 on a logarithmic probability axis (bottom). The dashed envelope is the maximum over the four tested widths; it has no one-standard-deviation interpretation. Values at $p_0 = 0.5$ correspond to no positive fitted excess.

The fully scaled/MC limit ratio has median 1.31 and ranges from 0.733 to 2.77. The minimum MC-scenario p_0 is 1.3×10^{-3} at 93 MeV; the fully scaled minimum is 2.9×10^{-3} at 66 MeV. These minima are local references, not probabilities for the full mass-and-width search. Orange shading identifies 90–100 MeV, where the 2015 scan is exploratory. The combined curves connect successive scan points through dataset transitions. Membership is 2015 at 19–38 MeV; 2015+2016 at 39–49 MeV; all three at 50–100 MeV; 2016+2021 at 101–180 MeV; and 2021 at 181–250 MeV. The common t is a coherent broadening path, not a determination that all calibration errors are correlated.

7 How the fitted uncertainty changes



Ratios to the MC scenario of the full-template fitted yield error, the RMS fractional GP predictive uncertainty in the extraction bins, and the signal yield per unit ϵ^2 . The GP diagnostic is not its covariance projected onto the signal direction.

For a known, locally flat background density β and a normalized Gaussian signal, the Fisher approximation for the fitted signal event yield N_s is

$$\text{Var}(\hat{N}_s) = 2\sqrt{\pi} \beta \sigma_m.$$

A 25% wider signal would then increase the yield error by $\sqrt{1.25} - 1 = 11.8\%$. That calculation omits GP uncertainty, changing training/extraction bins, and local excesses or deficits. These effects explain why the simple approximation is not a prediction of the observed curves.

The four-scenario envelope is $U_{\text{env}} = U_{\text{MC}} + \max_t (U_t - U_{\text{MC}}) \geq U_{\text{MC}}$. It is an upward variation summary on this finite grid, not a propagated nuisance result or a continuous maximum. An endpoint limit difference must not be added in quadrature with the fitted yield error: one is the nonlinear response of an interval, while the other measures fit curvature.

8 A one-sided width nuisance on a fixed experiment

The new nuisance calculation holds the observed bins, background mean b , GP covariance factor L , and yield conversion fixed while changing the bin-integrated signal shape. For each tested mass it uses the widest, fully scaled extraction mask and its selected GP prediction. The density-to-yield conversion is fixed at the MC value. Thus a width variation cannot change which data enter the likelihood or re-estimate the conversion.

$$\lambda_i = b_i + (L\theta)_i + AS_i(t), \quad \ell(A, \theta, t) = \sum_i \left[\lambda_i - n_i + n_i \log \frac{n_i}{\lambda_i} \right] + \frac{1}{2} \theta^T \theta + \frac{1}{2} t^2, \quad 0 \leq t \leq 1.$$

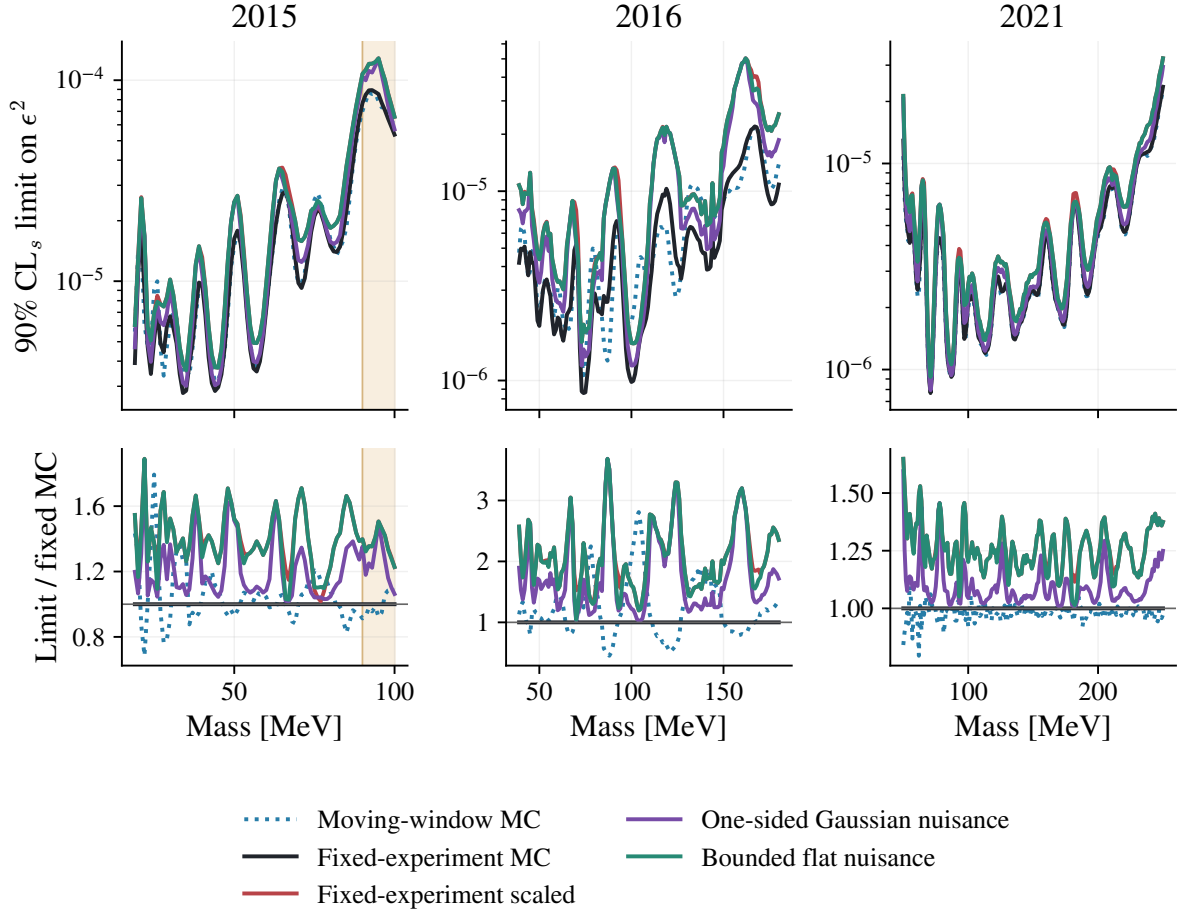
Here $A = \epsilon_{\text{native}}^2/10^{-8}$ and $S_i(t) = 10^{-8} K g_i(t)$, where K is the fixed full-template signal yield per unit native electron-channel $\epsilon_{\text{native}}^2$ from the MC-density conversion. Displayed limits apply the common dimuon factor after this fit. The exactly integrated Gaussian fractions $g_i(t)$ are normalized on the full stored support before restriction to the fixed fit bins. The constraint $\exp(-t^2/2)$ favors the MC baseline and treats the full correction as one constraint unit, with a hard endpoint at $t = 1$.

This is an assumed, truncated one-sided Gaussian constraint. A measured data/MC Møller ratio, such as 1.25, supplies a correction, but does not by itself supply its standard uncertainty or justify a hard allowed range. A calibration likelihood derived from the width measurement and its transfer uncertainty would be needed to assign that interpretation. The present construction tests the user's proposed MC-centered uncertainty model explicitly.

The width and background are profiled in both the tested-strength numerator and the physical likelihood-ratio denominator. The background-Asimov problem is profiled again at each tested strength; the observed envelope is not substituted for that calculation. The observed upper limit solves the corresponding nominal asymptotic $\text{CL}_s = 0.10$ equation.

Two same-experiment controls fix $t = 0$ and $t = 1$ while retaining exactly the same bins, GP constraint and MC conversion. These are the appropriate baselines for the nuisance result. They differ from the earlier moving-mask, reoptimized scenarios, even when the named signal width is the same. A bounded flat constraint over the same interval is included as a separately labeled modeling control.

9 Observed limits with the profiled width

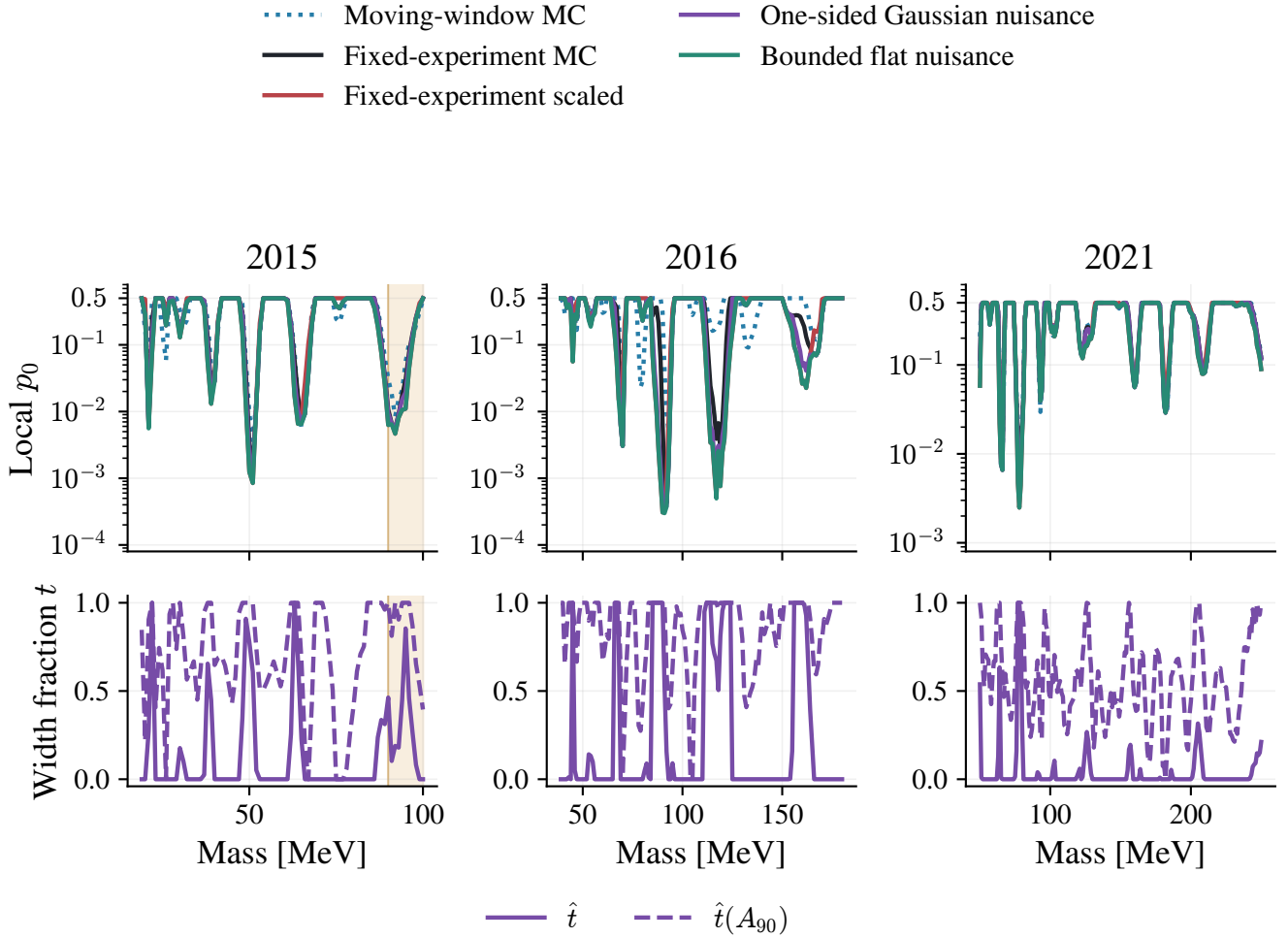


Individual-dataset limits with a continuous one-sided width nuisance (purple), a flat bounded control (green), and fixed MC/scaled widths on the same experiment (black/red). The dotted blue curve is the earlier moving-window MC result. Ratios use the fixed-experiment MC denominator; the blue curve therefore exposes the separate background/geometry change.

Dataset	Fixed MC / moving MC	Gaussian / fixed MC	Gaussian / moving MC	Flat / fixed MC
2015	0.976	1.15	1.14	1.38
2016	0.877	1.56	1.38	1.99
2021	1.02	1.08	1.09	1.25

Each entry is the median of the indicated pointwise ratio. The medians are computed separately and need not multiply. The Gaussian constraint is the primary uncertainty assumption; the flat bounded width gives its modeling control. The profiled curve is a nominal asymptotic interval under the stated constraint. Its displacement need not equal the four-scenario envelope or coincide with the fully scaled endpoint. No combined width-nuisance limit is inferred from individual results: a joint calculation must specify and profile the independent and shared calibration components.

10 Local references and fitted nuisance widths

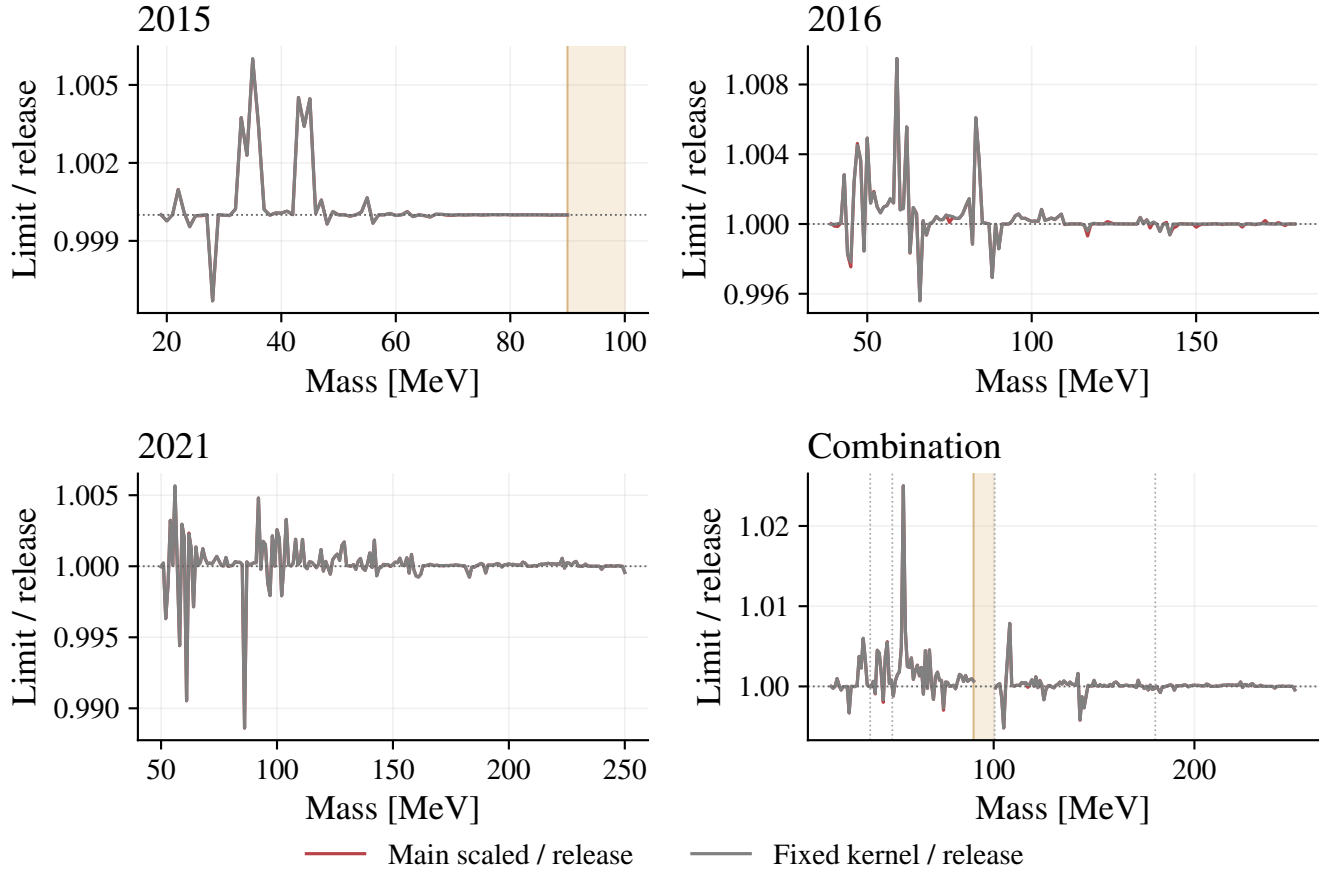


Nominal local p_0 transforms (top) and fitted width fractions for the one-sided nuisance (bottom). Solid $\hat{t}$ is the width at the fitted nonnegative signal strength; dashed $\hat{t}(A_{90})$ is the width at the upper-limit profile. Widths are constrained to the interval $0 \leq t \leq 1$.

For a fixed resolution scenario, the original reference is $p_0 = \Phi[-\max(0, r)]$, where r is the signed likelihood root and deficits give $p_0 = 0.5$. The displayed nuisance p_0 uses the analogous nominal transform of its profiled q_0 . It is not a newly calibrated discovery probability: the signal width enters through a component that vanishes at zero signal, and the assumed constraint and boundaries require care in applying asymptotic results.

Selecting the smallest reference over mass and width introduces an additional search dimension. No mass/width trials correction or null simulation is supplied here. A fitted width at a boundary does not measure a detector calibration shift. It describes how this constrained signal template accommodates the observed spectrum and the fixed background model.

11 Procedural controls at the fully scaled width



Fully scaled limits divided by the archived scaled release, comparing the main selected GP refits and the control with reviewed kernel parameters held fixed. Ratios are shown only where the archived and current dataset membership agrees. In particular, there is no archived all-three comparison for the new 91–100 MeV region.

A change in the moving exclusion reconditions the GP even when kernel hyperparameters are held fixed. That control therefore does not freeze the fitted background experiment. The main lane also permits hyperparameter optimization, with candidate selection based only on sideband marginal likelihood; no observed limit or p-value is used to choose the kernel.

The profile solver evaluates Poisson deviance and checks score residuals, avoiding cancellation from subtracting large absolute negative log likelihoods. Small differences from the archived release remain at unchanged width. They are procedural/numerical comparisons, not resolution uncertainty. The original random-start refits and fixed-kernel rows remain in the machine-readable ledger.

Above 90 MeV, the reviewed 2015 seed is extended from its 90 MeV value before scenario-specific refits. The underlying histogram support remains 14–135 MeV. This is a new exploratory procedure in that interval, not a replay of the older study’s frozen-kernel extension, and it carries no new support qualification.

12 Interpretation, source qualifications and reproducibility

Dataset	$t = 0$	$t = 0.4$	$t = 0.6$	$t = 1$
2015	3.7×10^{-3} (51)	1.6×10^{-3} (51)	2.0×10^{-3} (51)	8.5×10^{-4} (51)
2016	5.3×10^{-3} (93)	2.1×10^{-3} (92)	2.2×10^{-3} (92)	3.1×10^{-4} (90)
2021	6.9×10^{-3} (65)	5.8×10^{-3} (78)	4.6×10^{-3} (78)	2.5×10^{-3} (78)
Combination	1.3×10^{-3} (93)	1.8×10^{-3} (93)	2.5×10^{-3} (93)	2.9×10^{-3} (66)

Entries are minimum fixed-scenario local p_0 , with the mass in MeV in parentheses. Minima can occur at different masses. They must not be interpreted as global search probabilities or combined as independent observations.

Resolution sources. The inferred 2015 MC curve is the production curve divided by $1.03 \times (1.609/1.303)$. This removes the target systematic factor and rounded Møller correction; it is not a newly fitted raw-MC curve. For 2016, the raw and track-smeared Table 12 polynomials are interpolated pointwise. The 39 MeV edge inherits the source’s low-mass polynomial caveat; the 180 MeV upper edge avoids tail extrapolation. The 2021 source MC already includes Omega and hit smearing: unscaled removes the additional target-constrained Møller factor only. The exact production endpoint 1.25 is retained.

Claim boundaries. The scenario envelope is a finite-grid variation summary. The profiled one-sided nuisance is a real likelihood calculation under an assumed constraint. Neither supplies new coverage calibration, an expected-limit band, or a global discovery probability. The inherited 2016 qualification issue and exploratory 2015 extension remain explicit limitations.

Reproducibility. The package includes the scan rows, nuisance controls, source polynomials and audits, solver and report scripts, vector figures, generated L^AT_EX source, numerical checks and a checksum manifest. Native electron-channel limits are retained alongside the common display correction. The PDF is rendered page by page to check equations, labels, tables and the connected combined curves.

The complete main scenario scan has 2,628 rows. The two nuisance models give 850 limits at 425 individual mass coordinates. All completed independent row checks passed; 18 dense-width spot checks agree to a maximum relative limit difference of 4.21×10^{-10} . These are numerical and provenance checks, not coverage or discovery calibration.

References. [1] Particle Data Group, *Statistics* (2025), auxiliary measurements, [PDG review](#). [2] G. Cowan et al., *Asymptotic formulae for likelihood-based tests of new physics*, EPJC 71 (2011) 1554, [arXiv:1007.1727](#). [3] E. Gross and O. Vitells, *Trial factors for the look elsewhere effect in high energy physics*, [arXiv:1005.1891](#). [4] The pinned v5.3 resolution audit, statistics review and this continuation’s nuisance/extension validation records.